Pythagorean Theorem in Real-World Problems

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	6 — Relationships in Triangles
Lesson	Lesson 4 — Pythagorean Theorem in Real-World Problems	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.8.GR.1.1

2. Standards

MA.8.GR.1.1 Apply the Pythagorean Theorem to solve mathematical and real-world problems involving unknown side lengths in right triangles.

Must. Find a missing side in a real situation: the right angle located, the hypotenuse named, then the theorem; the answer in the story's unit, rounded once where it measures.

Must not. 3-D (a box's space diagonal — the book's 6.4.5); radicands over 225; coordinates (6.05–06).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.7.1	Apply mathematics to real-world contexts.	every board is a real object: a ladder, a ramp, a screen, a lawn, a kite — the triangle is found in it before any number
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	boards 7 and 9: the answer is checked against the story — does it fit, can it reach

4. Learning Target and Essential Question

Learning target. I can find a length in a real situation by finding the right triangle inside it.

Essential question. Where is the right angle in a ladder leaning on a wall — and why does that decide which length is c?

Building on. 6.03 — a missing hypotenuse or leg; 5.04 — real lengths from a plan; Grade 3–4 — a rectangle's corners are right angles.

Working toward. 6.05–06 — the same right triangle drawn on a grid between two points; Geometry — right-triangle trigonometry.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	11	Teach from the front. Expand once per law; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	10	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (10 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
A ladder's foot slides farther from the wall. Does the top go up or down? Why?	DOK 3
Why is the diagonal of a rectangle always longer than either of its sides?	DOK 2
A 10-ft ladder reaches 9.5 ft up a wall. Is its foot more or less than 3 ft from the wall? How can you tell without the theorem?	DOK 3

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	A tablet screen is 6 in wide and 8 in tall. How long is its diagonal?	***10 in***	14 — added the sides [MA.8.GR.1.1: the theorem is about squares]
2	A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall. How high up the wall does it reach?	***12 ft***	17.5 — added the squares: a 15-ft ladder cannot reach 17.5 ft [Notes III]
3	A ramp rises 1 ft over a run of 12 ft. How long is the ramp, to the nearest tenth of a foot?	***About 12.0 ft***	13 — added the rise and the run [MA.8.GR.1.1]; '12, the same as the run' — the ramp is the hypotenuse, longer [Notes I]
4	A 17-ft ladder reaches 15 ft up a wall. How far from the wall is its foot?	***A — 8 ft***	(B) added the squares: about 22.7 ft, farther than the ladder is long [Notes III: for a leg, subtract]; (C) subtracted the lengths, 17 − 15 [MA.8.GR.1.1: the theorem is about squares]; (D) stopped at $a^2 = 64$ and forgot the square root [Math Nation names skipping the final root]
5	A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner. How much shorter is Ava's walk?	***6 m shorter***	15 — Ava's walk, not the difference [the question asks how much shorter]; 21 — Ben's walk
6	A kite flies on 30 m of string, straight above a spot 18 m from where the string is held. Which length is the hypotenuse?	***C — the 30-m string***	(A) took the ground distance as c — it is a leg [Source of Truth, GR.1.1: misidentifying the hypotenuse]; (B) took the height as c — the height is the unknown leg [Source of Truth, GR.1.1: misidentifying the hypotenuse]; (D) did not see the right angle under the kite [Source of Truth, GR.1.1: failing to visualize the triangle in a real-world problem]
7	A box is 7 in wide and 12 in long. Will a 14-in umbrella lie flat in it, corner to corner?	***No — the diagonal is about 13.9 in.***	'Yes, 19 > 14' — added the sides [MA.8.GR.1.1]; 'Yes, 12 < 14' — compared with one side [Notes II: the diagonal is longest]
8	A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall. How high does it reach, to the nearest tenth of a foot?	***About 7.1 ft***	12.2 — added the squares (√149): higher than the ladder is long [Notes III]
9	A window is 12 ft above the ground. Why can an 11-ft ladder never reach it, wherever its foot stands?	***The height is a leg — always shorter than the 11-ft ladder.***	'Move the foot closer' — the height never passes 9 ft [Notes I: the ladder is c]

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of “convince the person next to you”, then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Every story starts with a sketch: the right angle marked first, then c. Boards 1–3 come with the sketch printed; the student labels it. $Leg^2 + leg^2 = hypotenuse^2$ stays on the board.

ELL. **Ladder**, **wall**, **ramp**, **diagonal**, **kite string**: point at the picture for each before the board. "How high" is a leg; "how long is the ladder" is the hypotenuse.

Enrichment. A 25-ft ladder stands 7 ft from a wall. Its foot slides 8 ft farther out. How far does the top slide down? (24 − 20 = 4 ft — both heights whole.)

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the ladder is the hypotenuse, the longest side, and the height it reaches is a leg — so an 11-ft ladder can never touch a 12-ft window, wherever its foot stands. A board that tries numbers has not named the hypotenuse.

Vocabulary. **diagonal**; **rise and run**

Materials. Whiteboards and markers. A meter stick or a ladder picture for Notes I if the room needs it. Calculators allowed, printed nowhere.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due at the start of the next class: Pythagorean theorem: word problems — 87U.